

DIRECTIONS *for questions 1 to 4:* The passage given below is accompanied by a set of **four** questions. Choose the best answer to each question.

Incubators, accelerators, garages, laboratories: the best big companies have had them for years. Whatever the moniker, in most cases a select few workers are liberated from the daily grind and encouraged to invent the future. Such innovation units are becoming *de rigueur* in the public sector too.

These government laboratories provide a bridge between the public and private sectors. Sometimes governments simply copy what private firms are doing. MindLab is based on the Future Center, the innovation unit of Skandia, a big insurance firm. Sometimes they get money and advice from private sources. Whatever the connection, these units plug the public sector into a new world. They are full of people talking about “disruption” and “iteration”.

The units also provide a connection with academia. Britain's Behavioural Insights Team was the world's first government outfit dedicated to applying the insights of behavioural economics to public policy. Its mission was to point out the “small details” of policy that can have big consequences. It persuaded HM Revenue and Customs, Britain's tax collection agency, to tweak the words of a routine letter to say that most people in the recipient's local area had already paid their taxes. As a result, payment rates increased by 5 percentage points.

Whatever their location, [government innovation labs] go about things in similar ways, with a lot of emphasis on harnessing technology. The most popular idea is co-creation – getting one's customers to help invent and improve products and services. ... Boston's Office of New Urban Mechanics has produced a series of apps which provide citizens with a convenient way of reporting problems such as graffiti and potholes (by taking a photograph and sending it to city hall, users provide it with evidence and GPS co-ordinates). ...

The most striking thing about these institutions is their willingness to experiment. Policymakers usually alternate between hostility to new ideas and determination to implement a new policy without bothering to try it out first. Innovation centres tend to be both more daring and happy to test things. ...

... Bureaucracies are designed to kill innovation in the name of predictability. And a change in the political wind can reverse sensible changes: New York's new mayor is undoing many of the reforms that the ex-mayor implemented. Politicians can also sometimes treat the existence of innovation centres as a justification for carrying on with business as usual everywhere. What is more, such centres spend a lot of time putting sticking plasters on cancers. It is admirable that France's Experimental Fund for youth provides young people with driving licenses at reduced cost. It would be better if the government changed the rules that made getting a license in France such a lengthy and expensive nightmare.

Still, something is better than nothing. And the various innovation centres have a growing number of somethings to their name. They have suggested some sensible policies [and] have also begun to change the institutional landscape. ... Reforming government is hard and often boring work. The innovation labs are making it a bit faster and a lot more interesting.

Q1. Which of the following sentences best captures the author's main viewpoint in the passage?

- ☐ a) The popularity of government laboratories, which connect the public with the private sectors and the academic world, has soared in recent times.
- ☐ b) Innovators and policy makers are at loggerheads with each other, and the former lose out to the latter more often than not.
- ☐ c) Innovation is the only hope for breaking away from the ineffective, antiquated ways of conducting businesses.
- ☐ d) Some bottlenecks notwithstanding, innovation centres can effect a positive change in the functioning of the government.

Q2. What is the style of the passage?

- ☐ a) Narrative
- ☐ b) Analytical
- ☐ c) Descriptive
- ☐ d) Argumentative

Q3. The phrase “putting sticking plasters on cancer” occurs in the penultimate para of the passage. By this phrase, the author implies that innovation centres

- ☐ a) always try to cover up a problem as they fear public’s wrath.
- ☒ **b) are looking at the correct problems but are failing to identify their root cause; this is hindering them from providing long-term solutions. ✓**
- ☐ c) value the public and hence want to find immediate solutions to the problems but they come up with somewhat less effective solutions.
- ☐ d) don’t want to spend much of their budget on innovation and hence try to find much cheaper solutions.

Q4. How would the “tweak” (referred to in para 3 of the passage) in the routine tax collection letter in Britain have prompted increased compliance?
It would have

- ☐ a) prompted the fear that the authorities would take legal action till taxes were paid.
- ☐ b) **reminded taxpayers of their obligations as citizens.**
- ☒ c) **shown that a small change in the content of the letter can have big consequences.** ✖ **Your answer is incorrect**
- ☐ d) encouraged the natural tendency of individuals to “go along with the group”.

DIRECTIONS for questions 5 to 8: The passage given below is accompanied by a set of **four** questions. Choose the best answer to each question.

For years, psychologist Michael Corballis, in his book *The Truth about Language*, has been one of the chief proponents of the idea that language has its origins in gestures.

Language is considered the jewel in the crown of human superiority over other animals. It emerges out of nowhere in our evolutionary past, and no other animal's communication comes close to it. Through clicks, hoots, barks and bleats, some animals can inform each other of how they are feeling. And some, through gestures, can signal anger, impatience, dominance or submission, desire and attachment. But none can use these displays to convey a sentence like “I kicked the ball”.

Being unique and powerful, human language is a siren to Darwinians, teasing us to examine how it arose, and from what. Corballis assigns a central role to the brain's mirror neurons, which seem to echo action in observation (in monkeys, mirror neurons fire when they reach to grasp objects, and when they observe another animal doing the same). There is an overlap between parts of the mirror-neuron system and two areas in the brain's left cerebral cortex that are associated in humans with language production – Broca's and Wernicke's areas.

Mirror neurons have their detractors but for Corballis they raise the possibility that language “evolved within a system that, back in our monkey days, was specialised for grasping things”. In humans, mirror neurons also seem to be involved with other actions. One is pointing. Conveniently for Corballis, this is one of the earliest milestones of communication in children, emerging around the first year, and signifying the beginning of shared attention. It is also disrupted in social disorders, such as autism. And the fine control of facial muscles required for speaking seems to share cortical circuitry with regions that control gestures.

Still, what is wrong with the view that our language grew from our hominid ancestors' capacity for vocalization? Charles Darwin thought that the capacity for complex vocal learning had deep evolutionary roots, extending at least as far back as our common ancestor with birds. Corballis counters that primate vocalizations, unlike gestures, seem barely under voluntary control. He contends that they emerge like nervous tics, more closely connected to expression of emotion than to a deliberate exchange of information. Vervet monkeys famously have three different alarm calls, corresponding to predatory eagles, leopards and snakes. But an alarm call does not involve the give and take of a language, gestural or otherwise.

One suspects that many – such as the primatologist and ethologist Frans de Waal – will find Corballis' assessment of primate vocalizations harsh. But primatologists such as Jane Goodall and David Premack share his views. ...

Q5. Which of the following choices correctly explains the relationship between the second and third paras?

- ☒ a) **The second para indicates the uniqueness of human language, and the third para identifies a possible cause.** ✓ Your answer is correct
- ☐ b) **The third para provides an explanation about the conclusion of the second para.**
- ☐ c) **The third para contradicts the content of the second para.**
- ☐ d) **The second para provides evidence that supports the view that the author refutes. The third para is part of the explanation that is used to support the main conclusion of the author.**

Q6. All of the following have been mentioned in the second para of the passage EXCEPT?

- ☐ a) **Language sets a human apart from an animal.**
- ☐ b) **Animal gestures cannot convey certain messages like “I kicked the ball.”**
- ☒ c) **Gestures are not sufficient as a source of communication and, hence, languages are also required.** ✖ **Your answer is incorrect**
- ☐ d) **Language is an instrument of action as much as a channel for the dispersal of information, while gestures help to express feelings in a powerful way.**

Q7. Which of the following choices correctly represents a feature or function of mirror neurons?

- ☒ a) **According to Corballis, mirror neurons are affiliated with language formulation in humans.** ✓ **Your answer is correct**
- ☐ b) **There is an overlap between the mirror neuron system and two areas in the brain's right cerebral cortex in humans.**
- ☐ c) **In children, mirror neurons help in making gestures for indicating their moods and help in intertwining language with their experiences.**
- ☐ d) **Mirror neurons fire when animals observe their human friends doing the same action.**

Q8. Which of the following aptly captures the essence of the statement “One suspects primate vocalisations harsh.” (Para 6)?

- ☐ a) **Frans de Waal agrees to disagree with Corballis.**
- ☐ b) **According to the author, Corballis’ analysis of Origins of Language is not acceptable.**
- ☒ c) **According to the author, Corballis’ assessment of communication in primates can be unacceptable to some primatologists.**
- ☐ d) Corballis says that primate vocalization is not as efficient a source of communication as language.

DIRECTIONS *for questions 9 to 12:* The passage given below is accompanied by a set of **four** questions. Choose the best answer to each question.

When lean mice become obese, neurons linked to feeding change their gene expression and activity levels. We know fatty, sugary foods can transform our waistlines. Much less is known about how they might transform the brain. Now, researchers have found that switching a mouse from standard chow to more fattening fare changes the activity of certain neurons that regulate eating, wearing out the cellular “brakes” that limit intake...

Changing an animal’s diet “is a subtle manipulation” ...The fact that you can go in and see different properties of these cells is an amazing finding. Neurobiologist Garret Stuber and a team at the University of North Carolina at Chapel Hill wanted to understand the brain changes that accompany obesity. They were particularly interested in an area at the bottom of the brain known to regulate feeding, called the lateral hypothalamus. With collaborators in the United Kingdom, Stuber – now at the University of Washington in Seattle – sequenced RNA from cells in this area of the mouse brain and grouped them according to what genes they expressed. By comparing gene expression between obese mice on a high-fat diet and control animals on a standard one, the researchers identified a subset of neurons that changed most dramatically with the obesity-inducing diet.

That set expressed the gene for an excitatory signalling molecule, glutamate. The researchers used a two-photon microscope to observe these so-called glutamatergic cells in the brains of living mice. They endowed the cells with a gene that made them fluoresce when they took up calcium – an indicator of neural firing. Then they watched the cells while mice lapped calorie-rich sugar water from a spout. If a lean mouse had just eaten, the neurons were more active in response to the sugar than if it had just been fasting. The cells seemed to act as a brake, signalling, “That’s enough!”

But as lean mice became obese from a diet high in fat and sugar, these cells became less lively. By 12 weeks after the diet switch, the glutamatergic cells were roughly 80% less active in response to the sugar drink...

The study is one of the first to use calcium imaging to observe brain activity in animals long-term... But the study doesn’t distinguish which changed the neural activity: a feature of the diet or the weight gain itself. It also doesn’t prove that the change in brain activity itself caused the animals to overeat and gain weight. The taste of the new diet might have prompted mice to up their calories – worn-out brake or no. Rodent chow [...] is dry, salty, bland grossness. High-fat mouse diets, on the other hand, usually taste like sugar cookie dough. Still, the reduced activity of the glutamatergic neurons might have perpetuated obesity in the mice by failing to tamp down their appetites even as their weight climbed.

...If future studies reveal a receptor present on this subset of neurons – but absent from most other cells – researchers could try to target it with drugs to selectively bump up activity. But that won’t be easy. Strongly stimulating these neurons is apparently unpleasant; mice in the study avoided electrical zaps to the area. So, the approach would require a delicate tap on the appetite brake, not slamming on it with full force.

Q9. Which of the following studies will add the least depth to the research presented in the passage?

- ☐ a) **A study to explore the presence of receptors which are present only on the subset of neurons that regulate eating.**
- ☐ b) **A study to explore whether the activity of glutamatergic neurons prevents overeating in lean mice.**
- ☐ c) **A study to explore the effect of increasing weight on the activity of glutamatergic neurons.**
- ☐ d) A study to explore the correlation between change in brain activity and overeating in the absence of obesity.

Q10. Which of the following best captures the reason for the author to feel that the approach would require 'a delicate tap on the appetite brake'?

- ☐ a) **Mice don't enjoy strong electrical zaps in the area where the neurons being studied are found.**
- ☐ b) **Bumping up activity of the receptors could help understand the exact function of the neurons.**
- ☐ c) **Over-stimulation of glutamatergic neurons is unpleasant to mice.**
- ☐ d) Only those receptors present on one subset of neurons and absent from other cells have to be targeted.

Q11. An experiment which establishes that the mice increase their calorie intake and weight only when fed with high-fat tasty mouse diets, and return to original levels of eating, weight and neuron activity when fed with rodent chow, will not show that

- ☐ a) it is not only neurons that control appetites.
- ☐ b) taste controls appetite more than our brain's internal mechanisms to prevent overeating.
- ☐ c) high-fat mouse diets lead to weight gain.
- ☐ d) overeating affects the activity of neurons that control appetite.

Q12. The author calls changing an animal's diet 'a subtle manipulation' because

- ☐ a) the diet change is done very slowly.
- ☐ b) it is a simple change to trigger change in the part of the brain the researchers want to study.
- ☐ c) the smallest change in diet can have huge consequential changes in the animal's brain.
- ☐ d) diet is the easiest thing to change in the animal's routine.

DIRECTIONS *for questions 13 to 16:* The passage given below is accompanied by a set of **four** questions. Choose the best answer to each question.

Deep in Europe’s manufacturing core, a German company is taking radical steps to cope with a perfect storm that has sent traditional economic pillars into a tailspin and put the country on the verge of recession. Huebner GmbH – a 73-year-old supplier of rubber and plastics that employs 3,300 people around the world – is selling its unit that makes accelerator pedals, keypads and auto-body parts after getting squeezed by Germany’s carmakers. Instead, it’ll focus on public transport and – in a completely new endeavour – making lasers.

The move reflects how Europe’s largest economy – reliant on exports and a world leader in the century-old industries of cars, machines and chemicals – is at a critical juncture. It’s being hit by [the] U.S. President’s protectionism and uncertainty [in Britain’s political scenario] just as companies struggle with a high-risk transition to greener products and smarter digital factories. The German economy shrank in the second quarter and could do so again this quarter... With day-to-day business rolled by geopolitical power plays, German giants like Siemens AG, BASF SE and Daimler AG have been forced to lower profit forecasts and seek ways to cut costs.

Smaller brethren like Huebner say they’re being pushed too hard in the fallout. The shift to new markets was made because big manufacturers are mistreating suppliers by “forcing them into ever lower prices,” said Reinhard Huebner, main shareholder, and son of the founder of the Kassel-based company...

Small and medium-sized enterprises (SME) such as Huebner make up the so-called Mittelstand, which accounts for 35% of corporate revenue and 58% of jobs in Germany. Often family-owned, they don’t have the cash buffers to sit out a downturn. Nor are they well-positioned to address structural challenges like a dramatic shift in the underpinnings of modern manufacturing. The SMEs are more-or-less sandwiched between big companies and their customers...

The ultimate challenge facing German manufacturing is the transition to so-called Industry 4.0, [the] fourth industrial revolution as manufacturers around the globe go digital. The aim is to combine robotics, machine learning, cloud computing, big data and the Internet of Things to create smart factories capable of better, more efficient production. While Germany originated the term, the country is ill-equipped to deal with the transition, with deficiencies in everything from software training to Internet infrastructure.

Amid complaints about a lack of government support and a shortage of trained labour, the Mittelstand is falling behind on research and development efforts. ... While political issues are distorting measures such as the exchange rate and raw material prices, making economic judgments “incalculable,” [these] companies still need to look further into the future...

For Huebner, that’s meant investing in lasers... The company’s products are now used for holographic films for augmented reality glasses and biomedical research. Sales have doubled since the unit was established [according to] Ingolf Cedra, Huebner’s managing director for materials and photonics. He’s optimistic that Germany [...] will ultimately weather the current turbulence. “German industry has a fantastic chance in combining the mechanical expertise, in designing the machinery and merging that with clever software solutions,” said Cedra. “I do believe that the Mittelstand – driven by machinery business, mechanical engineering – they have a chance, and they will take the chance.”

Q13. According to the passage, the ‘radical steps’ referred to in the first line of the passage do not include

- ☐ a) **a shift to making lasers.**
- ☒ b) **a shift to non-automobile markets.** ✖ Your answer is incorrect
- ☐ c) **closing down the rubber and plastics business.**
- ☐ d) **servicing public transportation and biomedical industries.**

Q14. All of the following are reasons behind the SMEs' bad shape in the fallout EXCEPT:

- ☐ a) **They don't have spare cash to last the downturn.**
- ☐ b) **It is a tremendous challenge to modify their manufacturing techniques.**
- ☐ c) **Their clients are trying to cut costs by pushing SMEs too hard.**
- ☐ d) They cannot sustain their business at the prices big manufacturers are setting.

Q15. Which of the following strengthens the author's statement that 'The ultimate challenge facing German manufacturing is the transition to so-called Industry 4.0'?

- ☐ a) **There are deficiencies in everything from software training to Internet infrastructure.**
- ☐ b) **There is a lack of government support and a shortage of trained labour for Mittelstand.**
- ☐ c) **There is an increased demand for greener products and smarter digital factories.**
- ☐ d) There is turbulence in exchange rate and raw material prices.

Q16. The perfect storm is most probably a metaphor for

- ☐ a) the traditional economic pillars of Germany which are going into a tailspin.
- ☐ b) the recession that is causing German economy to shrink.
- ☐ c) geopolitical and manufacturing shifts adversely affecting the German economy.
- ☐ d) competition for German giants that is squeezing their profit forecasts.

Q17. DIRECTIONS *for question 17:* The sentences given in the question, when properly sequenced, form a coherent paragraph. Each sentence is labelled with a number. Decide on the proper order for the four sentences and key in the sequence of four numbers as your answer.

1. Machine translation has never focused on understanding language.
2. Could it in fact be that understanding isn't needed in order to translate well?
3. Instead, the field has always tried to "decode" – to get away with not worrying about what understanding and meaning are.
4. The practical utility of Google Translate and similar technologies is undeniable, and probably a good thing overall, but there is still something deeply lacking in the approach, which is conveyed by a single word: understanding.

Q18. DIRECTIONS *for question 18:* The passage given below is followed by four summaries. Choose the option that best captures the essence of the passage.

Through its refined notions of what constitutes authenticity, postmodernism offers information professionals an orientation by which to consider the photographic process, to better appraise, preserve, and represent images. Understanding context and connotation improves the ability to provide deeper description of visual holdings and to make them more useful and usable to scholars. The traditional reliance of scholars on the written and spoken word has given way to photographs' being regarded as valid historical evidence. The information that images provide to their viewers and their capacity to evoke the past have earned them growing recognition as source materials for the study of history. Now, influenced both by an increased visual literacy and by a postmodern emphasis on the limitless circumstances of human actions, a greater understanding of photographs helps examine the context in which they have been created, used, and maintained in image collections.

- ☐ a) **Postmodernism can help scholarly research by enhancing the process of understanding and examining photographs for context and connotation.**
- ☐ b) **Scholarly research will benefit from the rise in credibility of photographs as historical evidence and the enhancement in the way photographs are analysed for context brought about by postmodernism.**
- ☐ c) **The historic information offered by photographs has been greatly refined by the lens of postmodernism allowing scholars to study them for context and information more thoroughly.**
- ☐ d) It is to postmodernism's credit that scholars are now able to use photographs as credible sources of historical context.

Q19. DIRECTIONS *for question 19:* There is a sentence that is missing in the paragraph below. Look at the paragraph and decide in which blank (option 1, 2, 3, or 4) the following sentence would best fit.

Sentence: If the first half is relatable, then the reader or listener will automatically make sense of the second half.

Paragraph: For antimetabole to be effective, it does not only have to be grammatically correct, but should also be logical. ____ (1) ____ People, after studying literature for a while, start thinking that they can churn out antimetaboles with a snap of a finger. ____ (2) ____ They fail to understand the fact that a sentence cannot be called an antimetabole if it is not based on a logical theme. ____ (3) ____ Antimetaboles are popular and effective solely because they appeal to reason and are easy to remember. ____ (4) ____ For example: “It is not about the years in your life, but about the life in your years.” A sentence like this can be called an antimetabole because it is appealing, correct (logically and grammatically) and has a message to convey to the readers.

- ☐ a) **Option 1**
- ☐ b) **Option 2**
- ☐ c) **Option 3**
- ☐ d) Option 4

Q20. DIRECTIONS *for questions 20 and 21:* The sentences given in the following question, when properly sequenced, form a coherent paragraph. Each sentence is labelled with a number. Decide on the proper order for the four sentences and key in the sequence of four numbers as your answer.

- 1. Or rather, they'll probably say 'webtoon' artist, that being the term of art for the form of comics now seen on screens all around the country.
- 2. Though digital, the format also harks back, if inadvertently, to the progenitor of all modern Korean comics: Old Man Gobau, whose four vertical panels appeared in national newspapers daily for 45 years, from not long after the Korean war until the final year of the 20th century, and whose creator Kim Seong-hwan died last month at the age of 86.
- 3. Start asking Korean high-school students, what career they want, and it won't be long before someone says they want to be a comic artist.
- 4. Unlike the horizontal newspaper comic strips, webtoons read vertically, from top to bottom, not for any reason to do with the now horizontally written Korean language but for better scrolling on a mobile phone.

Q21.

DIRECTIONS *for questions 20 and 21:* The sentences given in the following question, when properly sequenced, form a coherent paragraph. Each sentence is labelled with a number. Decide on the proper order for the four sentences and key in the sequence of four numbers as your answer.

1. But it should be a call to arms.
2. The act of imagining one's own children on the receiving end of unmitigated horror is, for most of us, just that – an act of the imagination.
3. If we instigate or participate in abuse, or if we turn a blind eye to it, how will we be remembered?
4. How will the smallest humans among us endure without a mother or father, or without adequate food, water, and shelter?

Q22. DIRECTIONS *for question 22:* There is a sentence that is missing in the paragraph below. Look at the paragraph and decide in which blank (option 1, 2, 3, or 4) the following sentence would best fit.

Sentence: To know something is one thing; but to know it with others, know that others know it, and know that they know that you know it, is another thing altogether.

Paragraph: One thing Kivas – a circular room with a bench formed out of straw and mud running around the perimeter – were used for is politics. When someone spoke in the Kiva, he could see his fellows and his fellows could see him. The circle also meant that his fellows could see each other seeing him; at a glance they could take in not only the speaker, but the faces of their colleagues doing the same. _____(1)_____ That matters because politics is not just what you think and believe. If I’m trying to do something that requires your cooperation, I need to do more than say I’m willing. I need to know that you know I’m willing. _____(2)_____ In cognitive science, we call this common knowledge. _____(3)_____ It’s what you need for costly cooperation. What teammates on the field and business partners in the boardroom signal when they look each other in the eye is a mental moment at the origin of society. Common knowledge is power. It is not enough to dislike a government; you need to know that others do too. _____(4)_____

- ☐ a) **Option 1**
- ☐ b) **Option 2**
- ☐ c) **Option 3**
- ☐ d) **Option 4**

Q23. DIRECTIONS *for questions 23 and 24:* The passage given below is followed by four summaries. Choose the option that best captures the essence of the passage.

There's a large body of medical literature showing that married people tend to be healthier and live longer than singles. But newer research adds an important caveat: the quality of the marriage matters. Surveys show that couples that enjoy good marital relationships are likely to share common interests when these shared interests have existed in the individual prior to the marriage. One should look for a spouse who shares common interests because marital stress, logically enough, is not good for one's health. Other studies have shown that happily married women have fewer blockages in their aortas, and that happily married couples are less likely than unhappy couples to suffer from heart disease.

- ☐ a) **Marital relation is a factor important enough to be considered in case of heart disease since the relationship between couples affects an individual's health and stress between them can lead to problems.**
- ☐ b) **As most heart diseases are due to stress and most of the stress is due to problems between spouses, marital stress needs to be controlled and considered in case of any heart disease.**
- ☐ c) **Happiness in marriage matters as marital stress is unhealthy, and happy marriages can be built by sharing of common interests.**
- ☐ d) Couples with shared interests are likely to have a happy marriage, and couples without shared interests develop heart diseases.

Q24. DIRECTIONS *for questions 23 and 24:* The passage given below is followed by four summaries. Choose the option that best captures the essence of the passage.

The global debate on climate change has become more realistic and nuanced. Gone are the old caricatures of bleeding-heart liberals on one end of the spectrum and staunch climate science deniers on the other end. Activists, academics and think tanks focused on climate action have been supplemented with heads of state, regulators and CEOs of some of the largest companies in the world who know they must at least publicly acknowledge climate concerns to pacify stakeholders. The sceptical end of the spectrum, however, is no longer the exclusive domain of unscrupulous business owners and unethical populists. There is no shortage of people with strong analytical backgrounds and mainstream reach arguing that we need to come to the sober realisation that the window of effective climate action has closed. This fatalism differs in tone from flat-out climate deniers. But the rationality of new climate sceptics makes their arguments more effective at squelching the will to fight climate change.

- ☐ a) **Despite high-profile parties now acknowledging climate concerns, rational individuals who now debate that fighting climate change is a lost cause are hurting chances of positive action.**
- ☐ b) **The climate change debate has become more complicated after a lot of well-known personalities have turned sceptical.**
- ☐ c) **With high profile people joining the climate change debate on either side, the issue of whether climate action is needed has become more nuanced.**
- ☐ d) Heads of state, regulators and CEOs have now taken the place of academicians and activists to argue against climate change deniers making the debate more realistic.